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Case Study (40%)

Case Study Title:

A Case Study of Alibaba Group

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MARKING FORM

TASKS		MARKS	
Report (70%)	Introduction		5
	Methodology		20
	Data Set		10
	Data Analysis and Results		20
	Conclusion		5
	Format		5
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Presentation (30%)	Clarity		10
	Presentation Slides		10
	Teamwork		10
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1.0 Introduction

1.1 Background of Alibaba Group

Jack Ma and 17 other founders launched Alibaba Group in Hangzhou, China, in the year 1999, as a basic online platform connecting Chinese manufacturers and international buyers via Alibaba.com(Alibaba Group, 2025). Over the subsequent decades, the Alibaba Group grew exponentially from this basic start to become one of the world's most complex and largest digital ecosystems.

Alibaba encompasses many segments that all tie back to its original premise of e-commerce, with numerous e-commerce platforms from its huge C2C and B2C businesses (**Taobao** and **Tmall**), along with global support to shop online from anywhere (**AliExpress & Lazada**). Additionally, Alibaba has invested heavily in technology through its digital payment service (**Alipay**) and its cloud service offering (**Alibaba Cloud**) (Brand Vision, 2025). It is expected that by the year 2025, Alibaba will reach over 1 billion customers and millions of small/medium-sized enterprises, controlling about 65% of China's online retail sales and approximately 60% of the packages being processed through Alibaba's logistics network as e-commerce packages in China. Alibaba's logistics processing ability is the highest in the world.

With its USD $25 billion IPO on the New York Stock Exchange in September 2014, Alibaba demonstrated its global reach and attracted international investors with a level of confidence that it had never before experienced in its history. The company's rapid growth stems from its long-term goal of "Making It Easy to Do Business Anywhere" and the establishment of its own vision of creating a company that will live beyond 100 years (Ma, 2017). Alibaba has been focused on a global growth strategy since 2014 and has expanded into 220 countries and territories around the world (Institutional Investor, 2014). Major investments internationally have included:

India: Major investments in Paytm, Big Basket, and Zomato (The Paypers, 2018)

Turkey and Pakistan: Significant acquisitions of Trendyol and Daraz(Alibaba Group, 2018).

Australia and Africa: Partnership initiatives to support cross-border trade and the development of small and medium-sized enterprises (SMEs).

In more recent times, the company has shifted its focus towards cutting-edge technology by investing heavily in the research and development of digital solutions to create an ecosystem for marketing and selling its products and services globally (what they call the “New Retail” concept that will bring together both the online and offline business models through data analytics and artificial intelligence. More specifically, the **DAMO Academy** and the **Tongyi Lab** have been established, as part of Alibaba's commitment to innovation, and will utilize the capabilities of AI to enable Alibaba to enhance its product and customer experience(Alibaba Group, n.d.).

Additionally, Alibaba's ability to leverage innovative algorithms and big data technologies like **MaxCompute** and **Flink** allows the company to process billions of events each day with a level of speed and efficiency that has never been seen before (Alibaba Cloud, n.d.).

1.2 The Significance of Exploring Retail Sales and Search Performance

The emergence of "New Retail," initiated by Alibaba, signifies the combination of traditional, physical-based businesses with internet, virtual-based businesses in the present day. A successful business model requires that retail performance be analyzed to ensure exporters have a means of survival or growth. In terms of size, Alibaba is an extremely large corporation with operations serving over 1 billion customers and accounting for over 60% of China's national e-commerce parcel delivery. Thus, the company's large number of transaction records, plus the vast number of other forms of data being collected throughout its operations, prohibit Alibaba from facilitating its operations through human oversight. Data Analytics provides the company with a digital nervous system, allowing it to take all the various and vast datasets collected within and outside the organization and identify the most relevant insights available to the company.

When analyzing performance metrics, companies must do so to accomplish specific goals.

- **Increasing Supply Chain Resilience:** Alibaba can turn the raw data collected throughout its supply chain into a set of resources to enhance a company's supply chain resilience and to respond more effectively to market or environmental challenges through the use of data network effects.

- **Enabling consumer engagement and personalization:** By analyzing the consumer's behavior through the clickstream and the search history, the company can develop deep learning algorithms (e.g., **Pailitao**) that link the user to the product with higher conversion rates (CVR).

- **Real-time operational agility** is exhibited by systems such as **RobustTAD**, enabling Alibaba to detect anomalies in both payment metrics and server metrics at response times under 100ms (Alibaba Cloud, n.d.), ensuring that large events such as the "Singles' Day" shopping festival remain operational without outages.

- **Strategic innovation:** comes from the application of data analysis to develop new business models, enhancing a company's ability to innovate quickly and successfully adapt to ever-evolving, unprecedented global occurrences.

1.3 Objectives of the study:

The objective of this study is to continue the practical aspects of the research that began in the first stage (Group Assignment). The primary objective of this phase is to link the theoretical foundations and findings from the first phase (literature review of Alibaba's analytics lifecycle) to the practical use of this information and the technical understanding gained through this research, using a real-world dataset.

Using a dataset from Alibaba's retail environment comprising 500 data points as the basis, the objective of this research is to demonstrate the entire data analytics process from data collection through data cleaning, modeling, and visualization.

The goal of this project is to demonstrate how analytical tools can contribute to achieving success for an organization, using data analytics to make the best business decisions possible based on data.

1.4 Objectives of the Analysis:

This study will use a "Big Question" to guide the analytical work. The "Big Question" relates to which variables in the dataset (search terms, demographics, and engagement metrics) best correlate with the objectives established in the study's first phase (Our Group Assignment).

The Big Questions:

"How do user search patterns and engagement levels (click-through rates) vary across different product categories, time periods, and demographic segments within Alibaba's digital ecosystem?"

Supporting Sub-questions:

- **Temporal Analysis:** How do search volumes and user engagement fluctuate over different time scales, specifically regarding the hour of the day and the day of the week?
- **Category Performance:** Which specific product search terms (e.g., designer clothing, electronics, food and beverage) generate the highest click-through rates and user interest?
- **Demographic Segmentation:** Are there significant differences in search behavior and engagement likelihood based on user age groups or gender?
- **Device & Location Impact:** How do different device types and geographic locations influence the way users interact with the search platform and their propensity to engage with products?

2. Exploratory Data Analysis:

The purpose of this section is to systematically analyze the dataset's foundational characteristics, i.e., the structural properties of the dataset, to gain a comprehensive understanding of the dataset and establish a foundation for more advanced analytic modeling.

	age	gender	country	search_term	hour_of_day	day_of_week	device_type	clicked
1	48	Female	Brazil	shoes	15	Sunday	desktop	0
2	32	Female	Sri Lanka	food and beverage	19	Thursday	desktop	0
3	31	Female	China	watches	24	Sunday	mobile	0
4	20	Female	USA	watches	2	Thursday	desktop	1
5	59	Female	USA	watches	22	Tuesday	desktop	1
6	60	Female	UK	jewelry	15	Monday	mobile	1
7	54	Male	Canada	food and beverage	17	Sunday	desktop	0
8	31	Female	China	home goods	11	Saturday	desktop	0
9	42	Female	China	shoes	11	Sunday	mobile	0
10	43	Female	China	designer clothing	6	Saturday	mobile	0

Figure 1 : Sample of the Dataset

2.1 Data Set Overview

The analyzed data set focuses on user search and transaction patterns within specific segments of the Alibaba Group. It includes detailed measures of engagement and interactivity with Alibaba's online retail services, such as Alibaba.com, Taobao, and Tmall, through the POS and search systems associated with each service.

- **Source of Data/Collection:** The data set was obtained from the 2019 search and transaction logs created by Alibaba (Rana, Alibaba Group, 2019) . The information is gathered directly from the Alibaba ecosystem's point-of-sale (POS) systems and search systems associated with each platform, including Taobao and Tmall, etc. The logs represent billions of events each day, providing a comprehensive understanding of Alibaba customers and Alibaba platform performance.
- **Time Frame:** The data set reflects all transactions conducted in 2019. This year is significant because it captures the increased use of AI technology to power Alibaba's various search systems and the company's efforts to expand globally at an unprecedented rate.
- **Number of Observations (Transactions):** Analyses were conducted using a data set containing 500 individual observations of behavior. This sample size enables detailed analysis of user trends segmented by both demographic characteristics and time.
- **Number of Variables:** The data set contains nine separate variables that provide a rich understanding of user engagement with Alibaba's retail platforms.

- **Demographics Used to Segment the Market:**

  - **Age** - Represents an important demographic variable for segmenting the target audience for each of the customized marketing strategies.

  - **Gender** - Is a category for the purchaser's respective purchasing preferences and behavior: female or male.

  - **Country (Location)** - Represents the user's country (e.g., China, USA, Brazil). It also serves as the reference point for Alibaba's operations worldwide (221+ countries) via its Cross-Border Trading Platform.

  - **Search Term** - Represents what the user is searching for. When a user does a search on Alibaba, the keyword(s) that make up the user's search are the search terms.

  - **Hour of Day & Day of Week** - Represents the time of day (hour) and day during which a transactional activity occurs, allowing for better systems management (operational handling) and volume management (asset management).

  - **Device Type** - Represents the type of device (mobile phone, desktop computer, or Tablet) from which the user interacts with Alibaba and through which they interact with the Alibaba platform.

  - **Clicked** - The posted metric of user engagement from the search activity enables the identification of the level of user engagement with the search results by recording either a history of the user clicking or not clicking on the product displayed on the Search Results Page (1 = click, 0 = no click). The user "clicked" metric represents the user's level of engagement with the search results.

2.2 Data Preprocessing and Quality Assessment

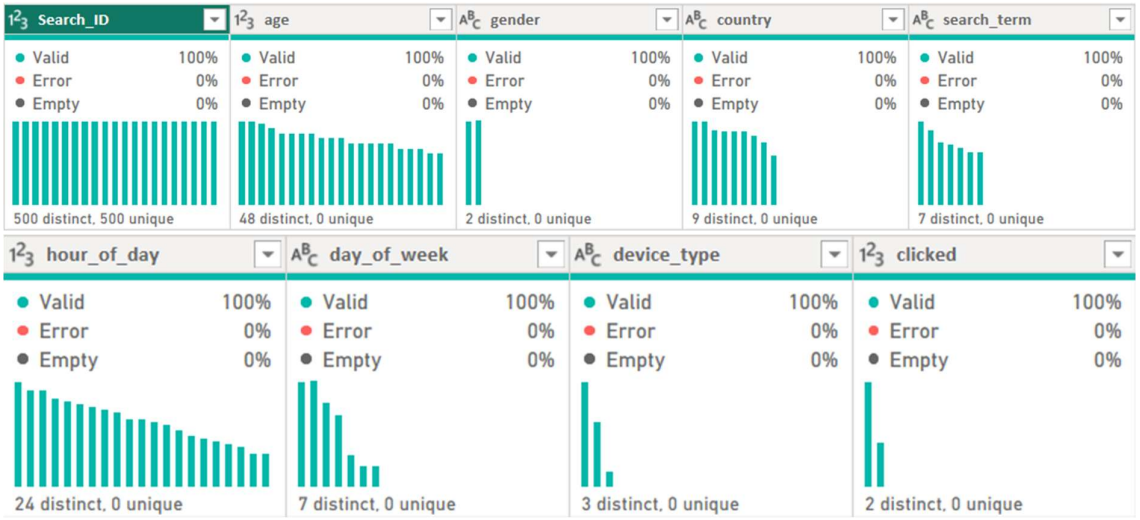

Figure 2: data validation

To guarantee the precision of analysis, a stringent data quality appraisal was carried out using Microsoft Power BI.To check against the Alibaba data's integrity, the follow-ing preprocessing steps were taken:

- **Attribute Renaming for Clarity:** We renamed the sequential index column to "Search_ID" so that each search transaction could be uniquely identified. This would mean analysis would not misunderstand individual query occurrences as different users, leaving a very clear primary key in the resulting dataset.
- **Missing Value Inspection:** Column Quality Feature in Power BI was used to inspect all aspects of the data. As demonstrated in **Figure 2**, the dataset is all-encompassing, with 100% of rows containing valid information and 0% missing or null values. It was unnecessary to use any imputation techniques to complete the required course construction.
- **Duplicate Removal Check:** search was made for duplicate records. This step was undertaken in order to try to avoid overestimations of search frequency. The dataset was cross-checked for identical rows of information – it transpired that no duplicate transactions existed, thus merely proving across-the-board that each record did indeed possess itself another moment from which it originated
- **Data Type Validation:** We reviewed the data schema to ensure that everything was correctly formatted. When loading, numerical fields (such as age) and text fields (eg. gender) were automatically determined and coded by Power BI. No manual type demarcations were therefore necessary

Conclusion on Data Quality: the dataset was found to be of a high grade and organizing stand. No heavy data cleaning or transforming was necessary in order to engage directly into exploratory analysis of these data.

2.3 Descriptive Statistics & Data Visualization

In order to find patterns in user behaviors, we conducted a descriptive analysis based on search trends, devices, as well as times.

Summary Statistics A high-level description of the data gives a perspective on just how many observations are in the data set that will be subjected to.

- **Total Search Volume:** There are 500 total unique search transactions in the dataset (Search_ID).
- **Metric Focus:** For the dataset in question, which is a clickstream dataset, the specific focus for analysis is the "Search Frequency" metric.

1. **Frequency Distribution by Device** The study examined devices that access the site in determining user preference. There is a distinction in usage patterns among devices as shown in **: Search** Volume by Device Type Search Volume by Device Type below:

   - **Observation:** Most searches come from Desktop devices, as 292 out of 500 total transactions came from Desktop devices. This is equivalent to 58.4% total traffic,

implying a much larger use of desktop interfaces compared to mobile/tablet interfaces.

- **Implication:** What is implied is that the interface as well as the marketing approach for the web platform should primarily cater to the Desktop users, so as to improve their user experience.

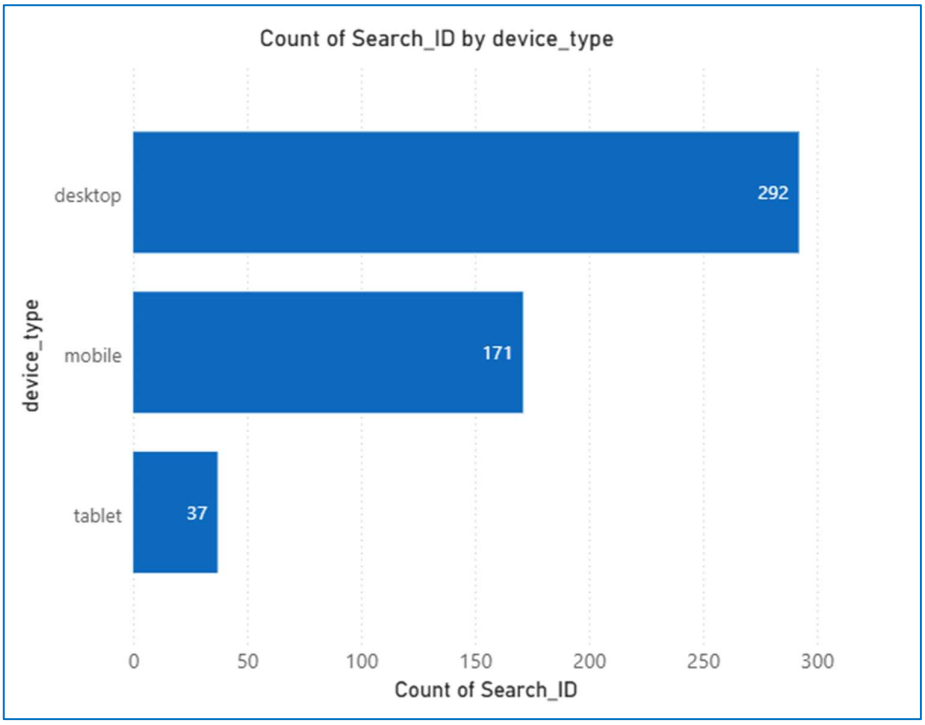

Figure 3: Search Volume by Device Type

Demographic Distribution To analyze the user base demographic, the gender distribution of the search queries was considered. The gender distribution of the search queries is presented in **: Gender** Distribution**.**

- **Observation:** The distribution is quite balanced. Female users comprise the dominant category, as they contributed 50.4% to the searches, while Male users contributed 49.6%

- **Implication:** The knowledge of this demographic divide is very important for formulating product recommendation and targeting strategies for the sueprior gender.

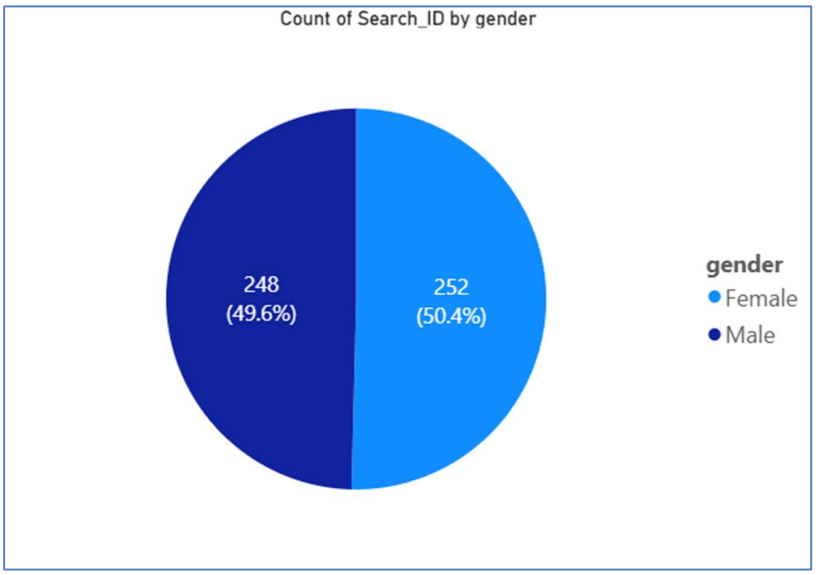

Figure 4 : Gender Distribution

2. **Temporal Pattern Analysis** To determine the times of the day when the searches are at their peak, the graph of the search volume versus the time of the day was used. **Figure 5** : Hourly Search Activitybelow shows the number of Search_ID's for each of the 24 hours of the day.

   - **Observation:** The lowest user activity occurs around 5:00 AM, probably during late-night sleeping hours. The activity steadily rises throughout the morning and reaches an absolute peak at 14:00 hours (2:00 PM).

   - **Insight:** This means that the usage is highest during the early part of the afternoon, and that is the best time to place advertisements or run flash sales.

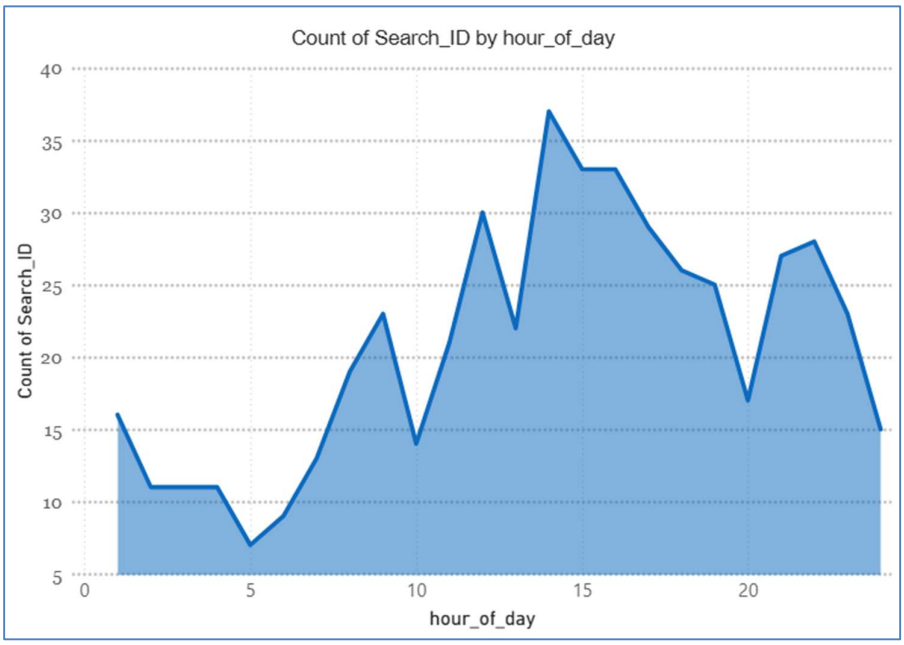

Figure 5 : Hourly Search Activity

3. Methodology

The following section describes how to analyze an Alibaba dataset through a systematic process. We will discuss the analytical methods and software that were used to evaluate the Alibaba dataset and to arrive at valuable results as they relate to the "Big Question." The methodology was developed to maintain the integrity of the data, as well as to provide a logical analysis of the differences in the way users search and engage with the content.

3.1 Tools and Software Used

For this case study, Microsoft Power BI was chosen as the main software for conducting data analysis. Throughout the entire analytics lifecycle of conducting the analysis, Power BI was used for the following activities:

- Data profiling to check the quality of the data columns and ensure that no values are missing and there are no duplicate records.
- Data modelling to set the type of each of the data (i.e., categorical versus numerical, example: Gender versus Age).
- Data visualization using interactive charts that display trends and comparisons.

3.2 Analytical Techniques

How to answer the "Big Question" and its supporting sub-questions? So to answer that, the following analytical techniques were applied:

1. Data Quality Assessment (Preprocessing) Before conducting an in-depth analysis of the datasets, a quality assessment of the dataset was conducted, assuring that the results would yield reliable data. The data quality assessment tool for this project was named "Column Quality Profiling" and consisted of a scan through all 500 observations to identify and mark any missing, incorrectly recorded, or duplicate records. As seen in **Figure 3**, all 500 of the original records were found to contain valid data, thus rendering imputation unnecessary.

2. Temporal Trend Analysis (Answering Sub-Question 1)

- **Technique**: Trend Analysis (Time Series).
- **Goal**: To understand how the total volume of searches and level of engagement changes over time.
- **Application**:
  - **Trends by Day of the Week:** Data was aggregated based on Clicks grouped by Day of the Week. A Line Chart (**Figure 6**) is used to illustrate how weekly engagement changes through the days of the week, revealing a growing trend towards increased engagement as we near the end of the week and the beginning of the weekend.
  - **Trends by Hourly View:** Total Click data was aggregated based on Hour of Day (0 - 23) to identify how many Clicks occurred, and how the amounts change throughout the week. An Area Chart (**Figure 7**), and not a Line Chart, illustrates the occurrence of peaks at 14:00 (2:00 PM) and valleys during the early morning hours. Areas and Lines were used as they show the trend of change continuously over time.

3. Comparative Analysis (Answering Sub-Question 2)

- **Technique**: Rankings and Comparative Analyses.
- **Goal**: To identify the type of product category that creates maximum user interaction and interest.
- **Application**: All clicks for each unique search term were summed, and this information was displayed on a horizontal bar chart (**Figure 8**) that ranks the product categories from highest to lowest. This allowed the user to compare the product categories that received the most traffic in a quick manner using the horizontal bar chart, e.g., "Designer Apparel" versus "Basic Necessities."

4. Descriptive Segmentation (Answering Sub-Question 3 & 4)

- **Technique**: Frequency Distribution and Demographic Segmentation.
- **Goal**: To understand who the users are by age, gender, and device type.
- **Application**:

- **Gender**: A pie chart was created (**Figure 4**) for visualization purposes to depict the ratio of male to female users. The pie chart indicated that there is almost an equal distribution of male and female users.
- **Age**: The age demographic of users was represented through the creation of a bar chart **(Figure 9**), which highlights the age range of users who performed the most searches on this site, specifically focusing on the age group of 50-60 years old.
- **Device Type**: The chart (**Figure 3**) depicts the relationship between the number of transactions generated by device type (desktop vs mobile) using a bar chart to compare the two by displaying the number of transactions performed using a desktop versus a mobile device.
- **Location** : the chart (**Figure 10**) depicts the relationship between type of device users and countries word wide displaying the geographic country and type device users

4. Data Analysis and Results

This section highlights the results that have been generated after conducting the analysis of 500 transactions. The analysis carried out includes trends, category sales, and demographic engagement to address the research objectives of the study.

4.1 Temporal Analysis (Weekly and Hourly Trends)

Analysis: Figure 6 below shows user engagement in terms of clicks on a weekly basis. There is a significant increase in engagement as the days go by, with a major spike on **Friday** with 35 clicks and another on **Thursday** and **Saturday** with 30 clicks each. The lowest level of engagement is observed on **Monday**, with an estimate of 6 clicks. Following this, one can assume that customers are most active during the days leading up to the weekend and least active during the early days of the week.

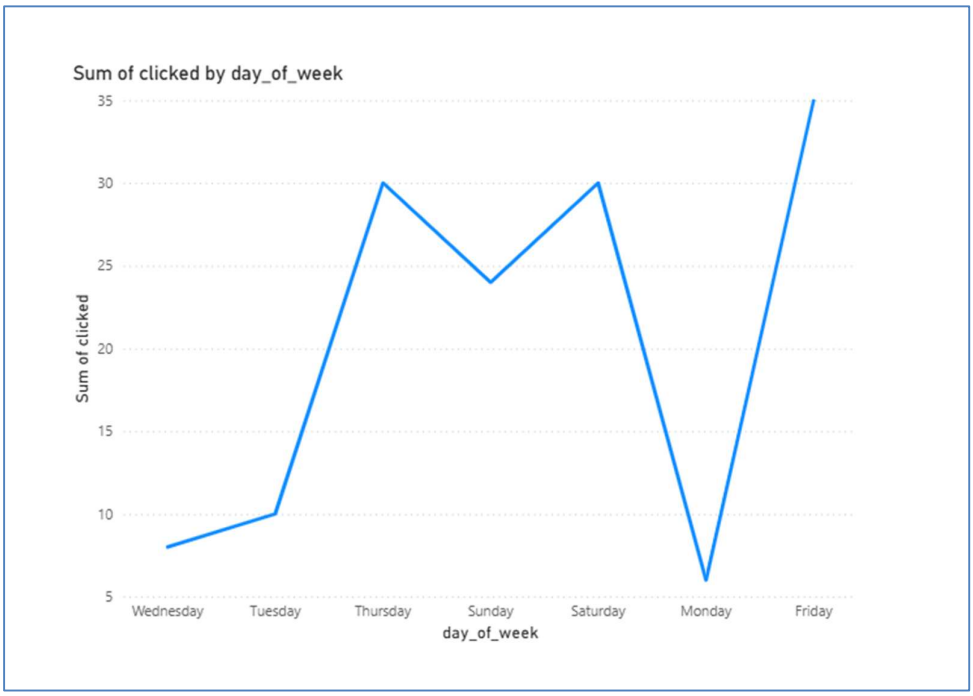

Figure 6: Weekly Engagement Trends

Analysis: Figure 7 shows that from an hourly perspective, there are fluctuations with regard to user activity. There is an enormous spike at **14:00 (2:00 PM)** to the point where it registers its daily peak with 14 clicks. The next peak occurs at **11:00.** However, there are low periods at 03:00 and 05:00 during the morning hours. This means that for this particular group, the shopping window occurs during the mid-afternoon hours.

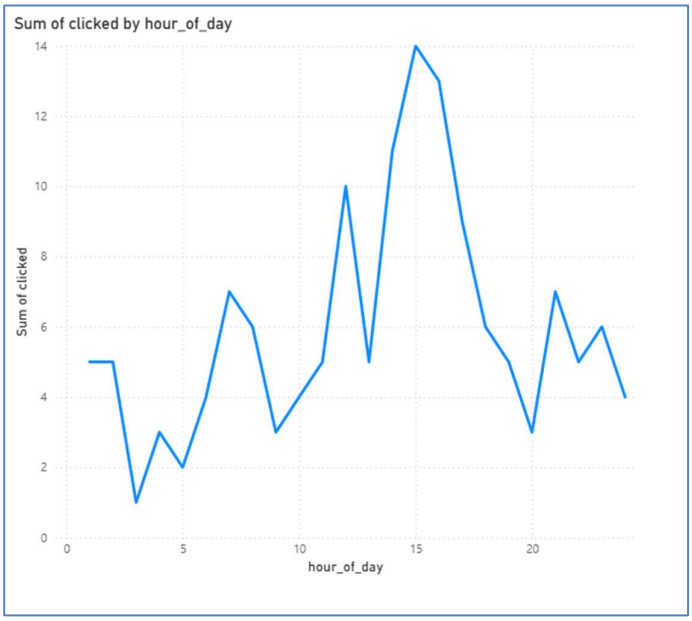

Figure 7: Hourly Engagement Trends

4.2 Category Performance Analysis

The graph in **Figure 8** below shows how different products have been classified according to the level of engagement they receive. The products that tend to create the highest level of engagement are Designer Clothing, which exceeds 30 clicks, along with Jewelry and Home Goods. It can also be observed that products like Food & Beverage and Clothes, which belong to the category that people normally require on a day-to-day basis, tend to offer the lowest engagement.

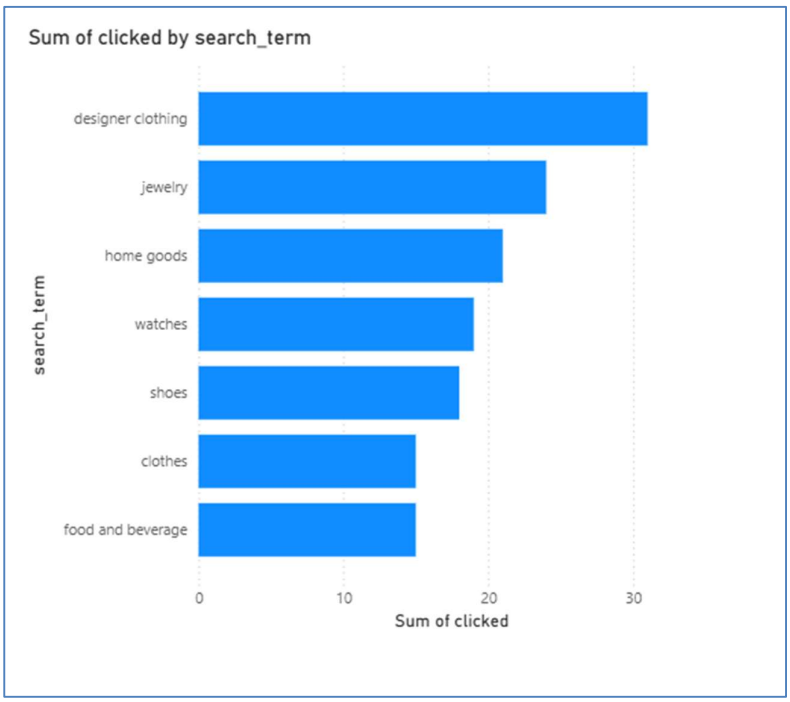

Figure 8: Engagement by Product Category Analysis

4.3 Demographic Analysis

Analysis: Figure 9 introduces the actors involved in the search process, indicating the search volume distribution by age categories. Interestingly, the older demographic does not leave the digital arena alone, especially those aged 50-60 years, contrary to the perception that the digital environment belongs to the young. These users comprise the 'silver economy' segment that represents the major proportion of the target audience, indicating that Alibaba provides an amiable platform even to the older age groups.

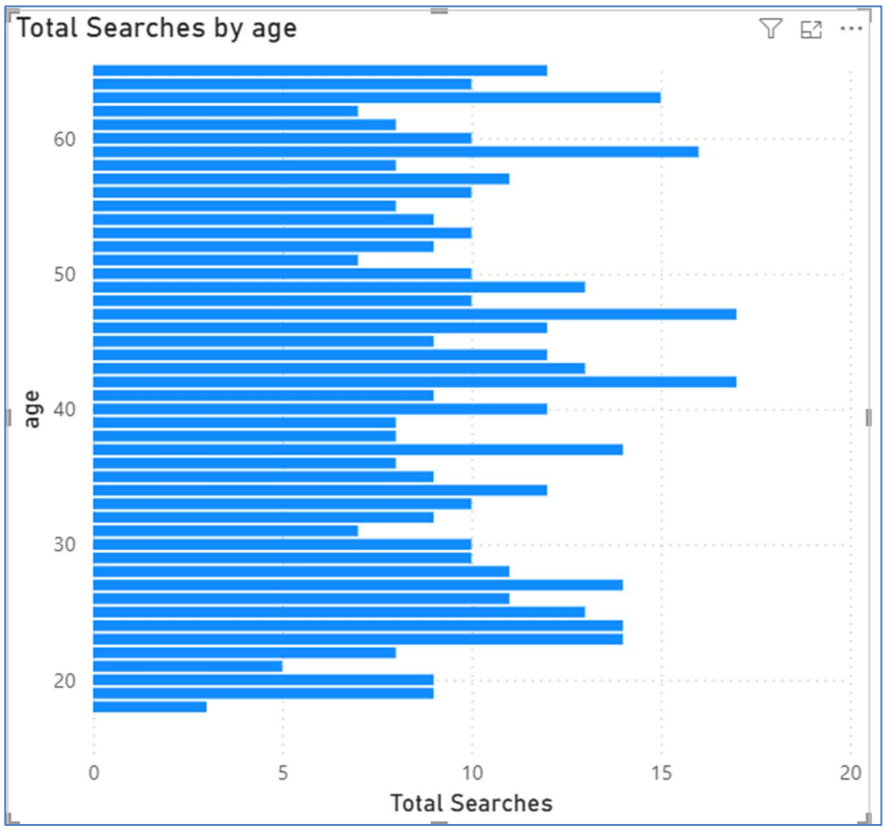

Figure 9: Search Activity by Age Group

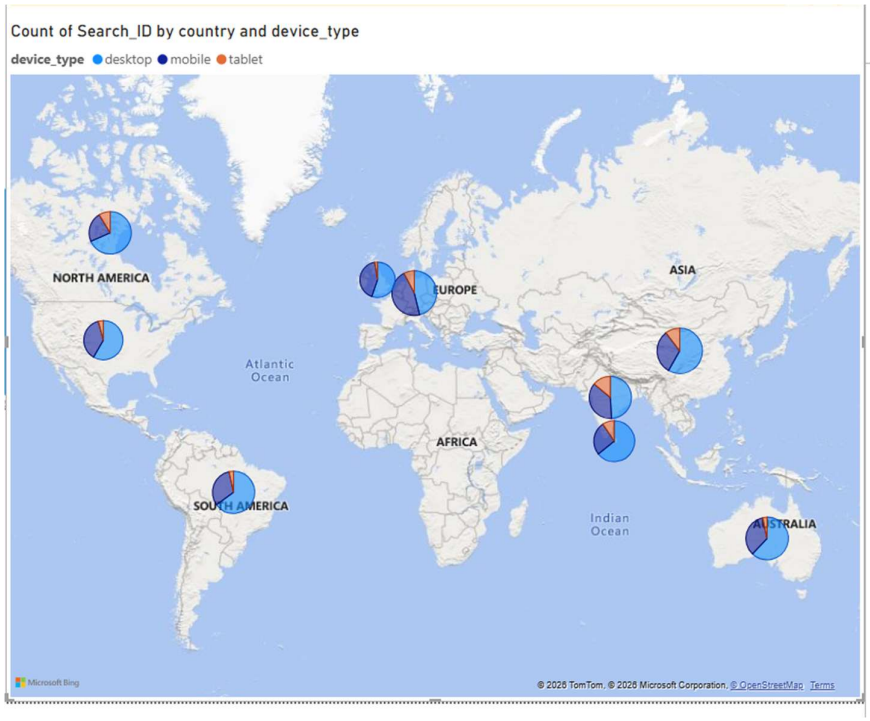

Figure 10 Count of Search_ID by country and device_type

As evident from the data represented in Figure 10, user engagement in the Alibaba universe has worldwide search engagement and reliance on specific hardware. The graph representing "Total Searches by Country" reveals search engagement in China with around 67 searches, followed by Germany with 65 searches and then Australia with 58 searches. There are prominent search engagements in Brazil, Canada, and India, that show a successful engagement of a broad worldwide user base in the Alibaba "New Retail" framework.

5.0 Discussion

Together, these elements combine the information to respond to the key Big Question and related sub questions posed by this study.

5.1 The Big Question

"How do user search patterns and engagement levels (click-through rates) vary across different product categories, time periods, and demographic segments within Alibaba's digital ecosystem?"

Answer: What the analysis shows: A highly contextual level of user engagement in the Alibaba ecosystem, driven by a particular "shopping window" (Friday afternoons), a particular type of luxury good (Designer Clothing and Jewelry), and a remarkably active mature user population (50+).
As opposed to a smooth distribution of user engagement, it appears timing and the level of luxury of a good are the two best indicators of purchasing intent.

5.2 Answers to Sub-Questions

1.Temporal Analysis: How do search volumes and user engagement fluctuate over different time scales, specifically regarding the hour of the day and the day of the week?

Week-long analysis shows the existing concept of having the "weekend shopping mentality," with the maximum number of clicks happening during Fridays and sustaining high levels throughout Saturdays. Daily analysis further identifies the exact timing of **14:00** or **2:00** PM as the favored engagement timing, and during early morning hours, no engagement is noticed.

2. Category Performance: Which specific product search terms (e.g., designer clothing, electronics, food and beverage) generate the highest click-through rates and user interest?

Clearly, consumers prefer luxury and lifestyle products over commodities. The categories that drove the highest levels of activity were "**Designer Clothing**" and "**Jewelry**," which outperformed more functional categories such as "**Food & Beverage**" by a large margin. As expected, users see this platform as a destination for making higher-end purchases.

3. Demographic Segmentation: Are there significant differences in search behavior and engagement likelihood based on user age groups or gender?

The site catered to a very mature demographic. The fact that the number of searches conducted by people aged 50-60 is very high challenges the perception that the purchase or use of e-commerce is the preserve of the young. Though the number of males and females is equal (about 50-50), the fact that the categories are fashion/jewelry are more important than gender segmentation.

4. Device & Location Impact: How do different device types and geographic locations influence the way users interact with the search platform and their propensity to engage with products?

The interaction is mainly driven by the worldwide preference for desktop computers that make it easier to search in a complex manner around the world. Although the country with the overall search volume is China with 67 searches, the page also receives a substantial amount of traffic from Germany with 65 searches and Australia with 58 searches, suggesting that the level of interaction is not restricted by the location or position of the users. The willingness for interaction is mainly influenced by the product type but not the location, and the interaction with the luxury product happens during the peak time at 14:00 hours.

6. Conclusion

To wrap up the research, user search patterns and engagement levels across time periods and product categories differ markedly. For example, “Designer Clothing” and “Jewelry” performed best, revealing a strong consumer preference for luxury items as opposed to necessities. At different times, engagement peaked at 14:00 (2:00 PM) and was greatest on Fridays pointed in the direction of a typical "weekend shopping mindset" for users. And as a side light, survey respondents over 50 years old were a sizable group--in fact, 58.4 percent of usage was via desktop computers.

Given these findings, Alibaba might choose to place an advance time-specific advertisement in line with peak usage on Friday afternoons and should improve its interface on Desk Top while customizing content for those older citizens. Nonetheless, the scope of this study was limited to a sample of 500 transactions chosen from a larger data set containing 100,000 records, and may not cover the full range of variability within its population.

For future, applying prediction models to the dataset would help make more accurate decisions. The methods that can be applied are: Logistic Regression, which can be used to predict the probability of a user clicking (0 or 1) or buying, using Age, Time, and Category as predictors. The second method can be Time Series Forecasting (ARIMA) to predict the number of searches for future events.

7. Strategic Recommendations

1. Meanwhile, Friday, 14:00 is just when being the highest traffic from users ideal time for running your marketing efforts and pushing ads. We should arrange high-value promotions and advertising businesses so that they coincide with the "weekend shopping psychology", that is for which we observe up to Friday at 12 o'clock (1-3pm). This period represents the absolute highest point of user engagement.

2. Focus on luxury and lifestyle goods. Our major market research indicates that users regard the platform as "a place to buy luxury goods." Investment in Designer Clothing, Jewelry and the like objects that perform exceptionally will pay off handsomely; however investment in functional categories like Food & Beverage is likely to bring only paltry returns for such a large outlay.

3. Optimizing for the "Silver Economy" is an imperative task. We must challenge the traditionally-heldview that e-commerce just serves younger people. Our core user has an average age from 50-60 years. This mature audience is the major cash (or marketing) cow at which we should aim without hesitation

4. Enhance the user experience on desktop. For desktop, while new media are indeed global, our statistics reveal that 58.4% of search traffic can still be classified as coming from desktop terminals. Therefore, we have to make sure that the working version should be fault-tolerant and highly user-friendly, especially for those people who work during morning and then browse during lunchtime.

5. Simultaneously Ever bridging descriptive analysis and prediction, this paper recommends that we should use Logistic Model in advance to predict the likelihood of users based on age and category. Also, we can predict the amount of search traffic per day for major events with ARIMA methods (Time Series Forecasting)—so ensuring enough server power and stock on hand.

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